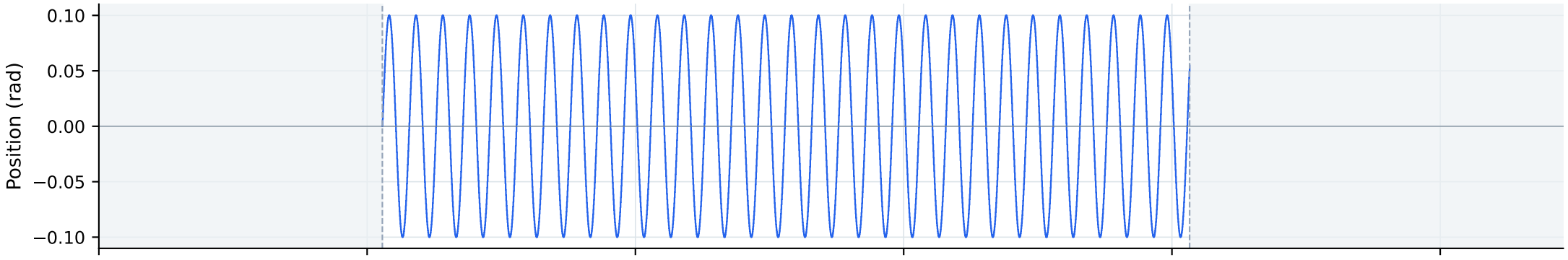
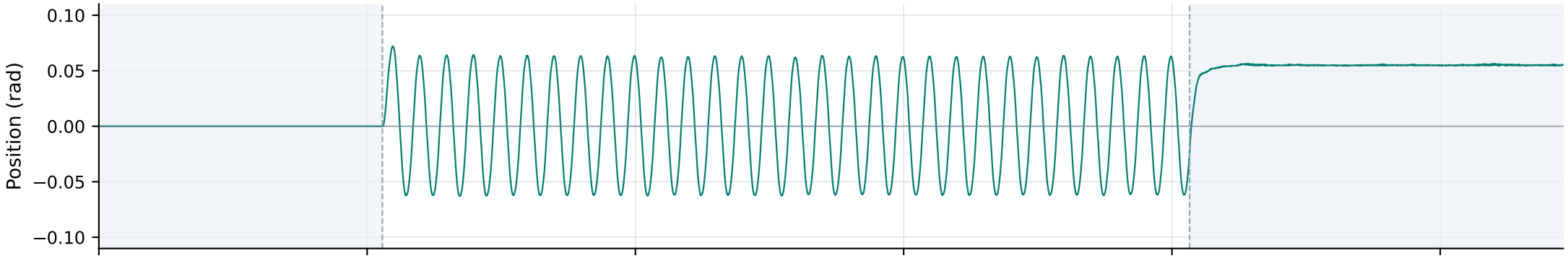
# Steering sine test — Full recording

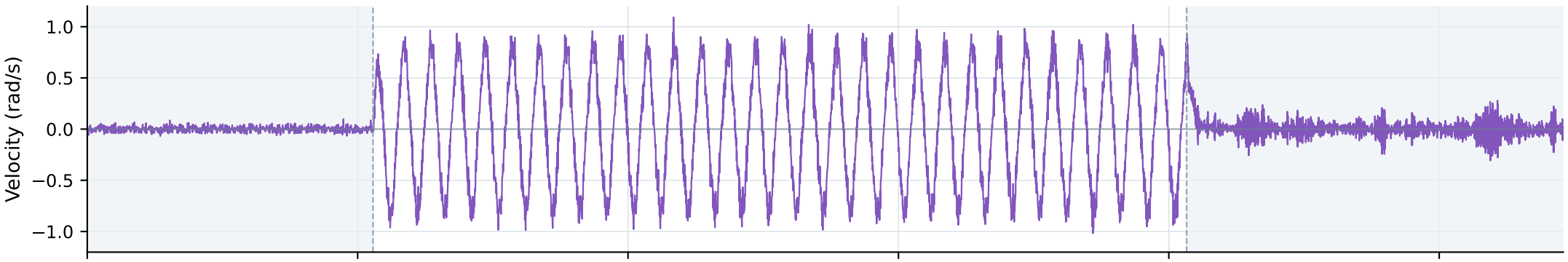
## 1 Steering position command · observed C2 targets



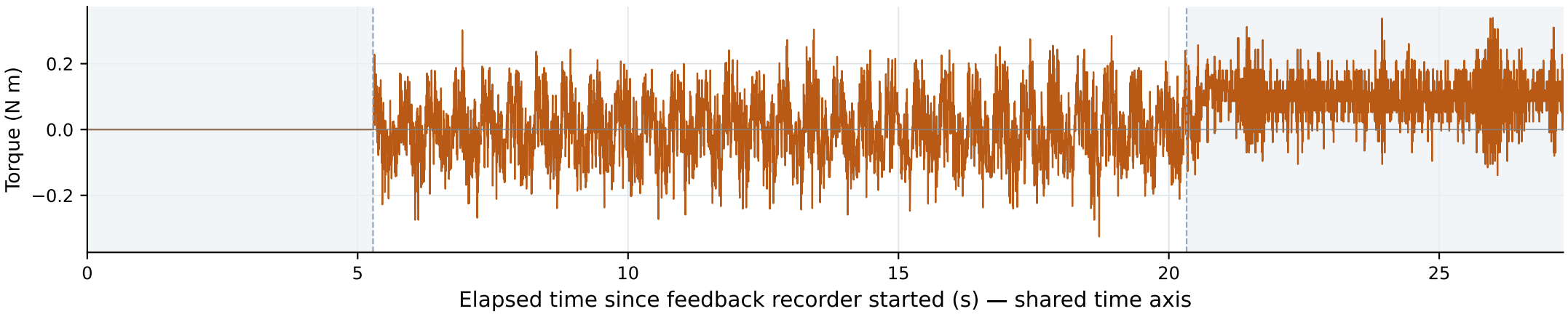

## 2 Steering position feedback · A3




## 3 Steering velocity feedback · A2




## 4 Steering torque feedback · A1 current × B0 constant




Collector startup A3 zero; positive = right. Grey = outside C2 interval. Red = feedback gap >50 ms. Torque = current-based estimate.